

Semiautomated feedback calibration instruction

for OPAT, ODC and DMA-22-05

Before any closed loop operation, the feedback input signal must be calibrated
It must be possible to move the valve-spool during the calibration steps. The movement is possible by handlever and/or electrohydraulic (hydraulic power needed) activation.

CAUTION/WARNING: With Hydraulic Power ON, the machine may move

Description (with Power/Enable on):

1. C1.00, and E 14 to "Off" / "0"
2. Set C1.09 depending feedback sensor type to "2" for Current feedback 4-20 mA (HPRS) OR "4" for Voltage feedback 0-10V (WA)
3. Open HCStool monitor (small "TV" icon) and select d1.11

TIP: Movement of spool by electrohydraulic activation via Setpoint (command) input is recommended as this confirm same sign (polarity) of setvalue and feedback input. If the signs are invers, the common cause is wrong/inverted wiring between valve twin solenoid and amplifier.

4. While moving the spool any direction, observe d1.11 falling or rising value. Spool direction showing:
 - Falling d1.11 value is defined as "Negative" side - verify d1.10 also negative / falling
 - Rising d1.11 value is defined as "Positive" side - verify d1.10 also positive / rising
5. Close monitor and, set C1.09 to the actual feedback signal type, 21-26. For this instruction only 21 (Bipolar voltage) or 24 (Bipolar Current) apply. Set C1.26 to "Off"
6. Open C1.39, select "Enter to re- or calibration of the feedback sensor" follow the below steps

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|----------------|---|--|
| Step 1: | Move spool to end stop Negative side . | <u>Click box 1</u> – wait for green checkmark |
| Step 2 and 3 : | Move spool to end stop Positive side . | <u>Click box 2</u> – wait for green checkmark
<u>Click box 3</u> – wait for green checkmark |
| Step 4: | Move spool to neutral (spring-centered). | <u>Click box 4</u> – wait for green checkmark |
| Step 5: | Move spool to end stop Negative side . | <u>Click box 5</u> – wait for green checkmark |

7. Message "Calibration Successful" appears, close C1.39 window
8. Open HCSTool monitor, select d1.11 and observe signal value 0 to -9.999V for Negative side and 0 to +9.999V for Positive side.
9. Set C1.26 to "1" or "2" and E 14 to desired value (normally 55 Hz)

The sensor is now calibrated and the closed loop regulator for the unit may be activated by setting C1.00 active, default is "1"

Observe the parameters affecting the closed loop controller before activating C1.00